

Piecewise Linear Manifolds

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Abstract

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1 Triangulated Manifolds

Definition 1.0.1 (Triangulated Manifolds)

Let M be a topological manifold. We say that M is a triangulated manifold if M is triangulated as a topological space.

Theorem 1.0.2 (Edwards)

Let $n \geq 5$. Let $\mathcal{K}$ be a simplicial complex of dimension n . Then $|\mathcal{K}|$ is a topological manifold of dimension n if and only if the following are true.

- $H_*(\text{lk}(\sigma)) \cong H_*(S^{n-|\sigma|})$ for all $\emptyset \neq \sigma \in \mathcal{K}$.
- $\text{lk}(v)$ is simply connected for all $v \in V(\mathcal{K})$.

1.1 Piecewise Linear Manifolds

Proposition 1.1.1

The set

$$\Gamma = \{f : U \rightarrow V \mid U, V \subseteq \mathbb{R}^n \text{ are open, } f \text{ is a PL homeomorphism}\}$$

defines a pseudogroup of transformations on $\mathbb{R}^n$.

Definition 1.1.2 (Piecewise Linear Manifolds)

A piecewise linear manifold is a manifold whose pseudogroup of transformations is given by Γ .

1.2 Combinatorial Manifolds

Definition 1.2.1 (Combinatorial Manifolds)

Let M be a topological manifold. We say that M is a combinatorial manifold if there exists a simplicial complex $\mathcal{K}$ and a homeomorphism $|\mathcal{K}| \cong X$ such that

$$\text{lk}(\sigma) \cong_{\text{PL}} \partial \Delta^{\dim(M)-|\sigma|+1} \quad \text{or} \quad \Delta^{\dim(M)-|\sigma|}$$

for all $\sigma \in \mathcal{K}$.

Lemma 1.2.2

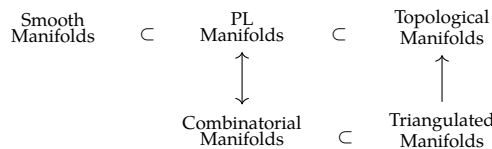
Let $\mathcal{K}$ be a simplicial complex. Suppose that there exists $n \in \mathbb{N}^+$ such that

$$\text{lk}(\sigma) \cong_{\text{PL}} \partial \Delta^{n-|\sigma|+1} \quad \text{or} \quad \Delta^{n-|\sigma|}$$

for all $\sigma \in \mathcal{K}$. Then $|\mathcal{K}|$ is a piecewise linear manifold of dimension n .

Theorem 1.2.3 (Whitehead 1940)

Let M be a piecewise linear manifold. Then M is a combinatorial manifold.



Note: in toric topology, PL manifold means combinatorial manifolds.

Theorem 1.2.4 (PL Hauptvermutung for combinatorial equivalence)

Let $\mathcal{K}$ and $\mathcal{L}$ be simplicial complexes such that $|\mathcal{K}|$ and $|\mathcal{L}|$ are combinatorial manifolds. Then $|\mathcal{K}| \cong_{\text{PL}} |\mathcal{L}|$ if and only if $\mathcal{K}$ and $\mathcal{L}$ are combinatorially equivalent.

manifold Hauptvermutung: if $\mathcal{K}$ and $\mathcal{L}$ are simplicial complexes such that $|\mathcal{K}|$ and $|\mathcal{L}|$ are homeomorphic triangulated manifold, then is $\mathcal{K}$ combinatorially equivalent to $\mathcal{L}$? This is proven to be false for dimension ≥ 5 .

1.3 Pseudo-Manifolds

Definition 1.3.1 (Pseudo-Manifolds From Simplicial Complexes)

Let $\mathcal{K}$ be a simplicial complex. We say that $\mathcal{K}$ is a pseudo-manifold of dimension n if the following are true.

- All maximal simplices of $\mathcal{K}$ are of dimension n .
- Each $(n-1)$ -simplex of $\mathcal{K}$ is the face of two n -simplices of $\mathcal{K}$.
- For any two n -simplices $\sigma, \sigma' \in \mathcal{K}$, there exists n -simplices $\sigma = \sigma_0, \dots, \sigma_k = \sigma'$ such that $\sigma_i \cap \sigma_{i+1}$ is an $(n-1)$ -simplex for $0 \leq i \leq k-1$.

Definition 1.3.2 (Pseudo-Manifolds in General)

Let X be a space. We say that X is a pseudo-manifold if X is triangulated by a simplicial complex $\mathcal{K}$ that is a pseudo-manifold.

Lemma 1.3.3

Let X be a pseudo-manifold of dimension n . Then $H_n(X) \cong \mathbb{Z}$ or 0 .

Lemma 1.3.4

Let $\mathcal{K}$ be a triangulation of S^n . Then the following are true.

- $\mathcal{K}$ is a pseudo-manifold.
- If $\mathcal{K}$ is finite, then the generator of $H_n(\mathcal{K})$ is represented by the sum of n -simplices of $\mathcal{K}$.